

Unsupervised Learning Project: A Reproducible Experimental Study of Clustering

Project Template: Milestone 2

Recommended format: 2–4 pages plus appendices/screenshots if needed. Conciseness and clear justification of choices will be valued.

Front matter

Team	[team Group / students names and numbers]
Project title	[title]
Checkpoint date	30 April 2026
Repository	[GitHub URL]
Repository version shown in lab	[commit hash / tag / branch]

1. RQ1 and segmentation time (very short)

- State the concrete segmentation question in 2–4 lines.
- State the unit of analysis: one row = one booking record.
- State the segmentation time (index time) assumed for clustering inputs.

2. Dataset version control and documentation

- Course dataset release/version used.
- Dataset file fingerprint (checksum/hash) recorded for version identification.
- Source / reference / usage terms noted: [yes/no + short note]
- Known data-quality issues already identified: [2–5 bullets]

3. Completed Exploratory Data Analysis (EDA)

- Provide a numerical missingness summary by attribute type.
- Provide a short outlier summary for the numerical variables considered relevant.
- State any variables dropped before modelling and why.
- Report at least the mandatory leakage exclusions (e.g., `is_canceled`, `reservation_status`) and any additional exclusions justified.

4. Main representation (RQ2)

- List included variables.
- List excluded / profiling-only variables and justify them.
- Describe numerical preprocessing: imputation, transforms, scaling.
- Describe categorical preprocessing: rare-category rule, one-hot/cumulative binary encoding, unknown-category handling.
- State the final `representation_id`.

5. Baseline modelling evidence

- Recommended at this checkpoint: k-means + iK-means.
- State the fixed K values considered and why they are plausible.
- Report internal indices computed in the same representation/metric space used for clustering.
- Report runtime for K-means/iK-means methods.
- Provide a first cluster-interpretation note or a very short profile summary.

6. Sensitivity / robustness plan already defined

- State one main representation
- State the planned seeds or resampling rule (minimum 10 runs where randomness applies).
- State the planned stability measure (e.g., ARI between runs).
- State clearly what question the sensitivity comparison is intended to answer.

7. Experiment logging and repository status

- Show that experiments.csv already exists or provide its fixed schema.
- Recommended columns: date/run_id, representation_id, method, parameters, seed, sample rule, metrics, diagnostics, notes.
- State whether environment.yml / lockfile already exists.
- State whether a run command or entry point already executes the current pipeline.

Suggested Appendix (screenshots/tables)

- feature-selection table;
- ColumnTransformer / Pipeline sketch;
- missingness table and outlier table;
- baseline-results table (k-means and iK-means);
- experiments.csv excerpt;
- one initial cluster-profile table or figure.